

Modelling tropical cyclone over-water wind and pressure fields

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Abstract

A parametric wind field model and an interactive computer tool are used to develop time histories of 2D wind field (surface wind speed and direction) during the 64 historical tropical cyclones that threatened the east coast of Queensland, Australia, during the 33 years (1969–2002) after the advent of meteorological satellite data for the region. These wind fields are suitable to be applied to applications such as wave, storm surge and circulation modelling. The parametric wind field model includes a double vortex and a representation of the synoptic winds in which the tropical cyclone is embedded. The process used all easily accessible meteorological data, including wind speeds, directions and pressures to determine the best-fit wind fields. Detailed comparisons between model and measurements for three individual storms are shown. These highlight the difficulties associated with fitting the parametric model to the very sparse measurements. If the storm did not pass close to a measurement station, then estimates of parameter values for radius of the storm and spatial shape of the wind field make the calibration process difficult and subjective. Nevertheless, results from the whole ensemble show that the modelled wind fields are unbiased with very good error statistics. These calibrations provide information for the distributions of values for radius of maximum winds and Holland B , parameters for which data are nonexistent in the Coral Sea region. An example of the use of the calibrated wind fields for the forcing of storm surge and wave models shows a very accurate comparison between modelled and measured surge and waves.

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1. Introduction

Parametric tropical cyclone wind field models (e.g. [Holland, 1980](#); [Harper, 1999](#)) are important for providing forcing for numerical models of storm surge and waves, especially for studies (e.g. [Hardy et al., 2003](#)) that require modelling a very large number of storms in order to determine the statistics of the effects of the tropical cyclone population. Studies have been conducted (e.g. [Shea and Gray, 1973](#); [Atkinson and Holliday, 1977](#); [Samsury and Zipser, 1995](#)), designed to enhance understanding of tropical cyclone structure and dynamics and have been used to improve the characteristics of these parametric wind field models. However, most of these studies have been undertaken for the North Atlantic and Western North Pacific cyclone basins. One particular on-going program is the cyclone fly-through program conducted in the US that has enabled researchers ([Samsury and Zipser, 1995](#); [Weatherford and Gray, 1988](#)) to develop horizontal profiles of cyclone winds that are important for determining the size and shape of tropical cyclone wind fields. Unfortunately, data are very limited in the Australian region where only satellite since 1969 ([Holland, 1981](#)) and radar (when the cyclone is within range) are available to potentially provide any indication of the size (i.e. the radius to maximum winds) of a storm.

We present the results from a study undertaken to develop 2D wind fields for 64 historical tropical cyclones (Appendix A) that have affected the east coast of Australia since 1969. These results are developed using a parametric tropical cyclone wind field model CycWind (cyclone winds) based on the parametric model presented by [Holland \(1980\)](#). A specially designed interactive graphical calibration tool has been instrumental in obtaining the best fit to available data. The result is a database of wind fields spanning 33 years (1969–2002) of east-coast cyclone activity. These wind fields can be used directly or applied to various applications including storm surge, wave, and circulation modelling that can be further applied to biological and physical studies along the Queensland East Coast and within the Great Barrier Reef World Heritage Area.

An additional outcome from this calibration process is the distribution of the wind field model parameters (radius of maximum winds, Holland B shape parameter). These too are important since they are required for undertaking Monte Carlo style simulations to develop long-term wind statistics such as those presented by [Gomes and Vickery \(1976\)](#) and recently by [Harper \(1999\)](#). There has previously been no other attempt to quantify the form of these distributions specifically for the Queensland coast because of the paucity of observations. These past studies have adapted distributions of parameters formulated using data from other tropical cyclone basins around the world.

2. Wind field model description

CycWind is a double vortex tropical cyclone pressure and wind field model that incorporates a synoptic scale wind field capability. It is an extended [Holland \(1980\)](#)

based model. The pressure profile is based on Cardone et al. (1994), which has a primary and secondary cyclone pressure profile specification. The pressure profile is used to determine gradient level wind speeds and directions. A synoptic scale wind field is merged into the cyclone wind field at gradient level and a boundary layer correction is applied to give wind speeds and directions at the surface (10 m above sea level). In addition, a first order wind field asymmetry is applied to account for the forward movement of the cyclone vortex.

The extended pressure field as described by Cardone et al. (1994) is calculated using

$$p = p_0 + \sum_{i=1}^2 \Delta p_i e^{-(R_i/r)^{B_i}} \quad (1)$$

$$\sum_{i=1}^2 \Delta p_i = p_\infty - p_0$$

where p is the pressure at radial distance r , p_0 is the central pressure, p_∞ is the ambient pressure, Δp_1 and Δp_2 are the primary and secondary vortex pressure deficits, R_1 and R_2 are the radii of the primary and secondary vortices, and B_1 and B_2 are the primary and secondary Holland B parameters. One method to relate B and B_0 is to use the empirical formula:

$$B = B_0 - \frac{p_0}{160} \quad (2)$$

The gradient wind speed, V_{gc} , due to the cyclone, which is a balance between the pressure gradient force, the Coriolis force and the centrifugal force, is calculated using

$$V_{gc} = \sqrt{\sum_{i=1}^2 V_{ci}^2 + \left(\frac{rf}{2}\right)^2} - \frac{r|f|}{2}, \quad (3)$$

where the cyclostrophic wind speed is given as

$$V_{ci} = \sqrt{\left(\frac{R_i}{r}\right)^{B_i} \frac{B_i \Delta p_i}{\rho_a} e^{-(R_i/r)^{B_i}}}. \quad (4)$$

This profile is used to simulate not only concentric eye-wall cyclones (as in Cardone et al., 1994), but also to improve the fit of the model wind speeds to those measured more than $3R_1$ from the centre of the cyclone. The adoption of the Cardone pressure profile formulation was motivated primarily by the existence of monsoon troughs in which cyclones in the Coral Sea off Queensland are often embedded. The formulation has worked very well in enabling better fits in a wide range of tropical cyclones in both the Coral Sea and Australian Indian Ocean.

Additionally, a synoptic scale wind field can be merged with the cyclone wind field. This has been included to simulate the interaction between the cyclone vortex and mid-latitude high-pressure systems that are commonly present during Coral

Sea tropical cyclones (Callaghan, 1996). This interaction can cause steep atmospheric pressure gradients that result in strong winds over a large fetch. The gradient wind velocity for the combination of cyclone and synoptic wind is given by

$$V_g = V_{gs} + V_{gc} \quad (5)$$

where the synoptic gradient wind is

$$V_{gs} = k_3(b_1k_1 + b_2k_2)V_s \quad (6)$$

with V_s an input value of synoptic gradient wind in m/s and

$$b_1 = b_2 = 0.5 \quad (7)$$

$$k_1 = \frac{V_{gcm} - V_{gc}}{V_{gcm}} \quad r > R_1$$

$$k_1 = 0 \quad r < R_1 \quad (8)$$

$$k_2 = \frac{V_s - V_{gc}}{V_s} \quad r > R_1, \quad V_s > V_{gc}$$

$$k_2 = 0 \quad r < R_1, \quad V_s < V_{gc} \quad (9)$$

in which V_{gcm} is the maximum gradient wind speed within the primary cyclone wind profile. These equations were designed to ensure smooth blending between the synoptic and cyclone wind fields. The variable k_3 is a simple weighting function to reduce the contribution of the synoptic wind on the equator side of the cyclone. It has a value of one on the southern side of the cyclone and is linearly reduced from one at the cyclone to zero 10° N of the cyclone.

The correction for the forward motion of the cyclone has been improved over previous work (McConochie et al., 1999) for use in this study. The forward speed correction is reduced with increased radius from the centre of the cyclone using the ratio of the gradient wind velocity (with no forward motion correction) and the maximum gradient wind velocity (with no forward motion correction). The gradient wind speed is

$$V_{gr} = V_g + \alpha V_f \quad (10)$$

where

$$\alpha = \frac{1}{2} [1 + \cos(\theta_f - \theta_{\max})] \frac{V_f}{V_{gcm}} \quad (11)$$

where V_f is the storm forward speed, θ_f is the storm direction, and $\theta_{\max}$ is the angle of the maximum wind speed in the cyclone, relative to θ_f . It is typically selected to be between 45° and 135° from θ_f .

In order to obtain the surface winds, a boundary layer wind speed and direction (inflow) correction is applied to the gradient wind. The surface wind speed at a 10 m elevation is calculated using

$$V_{10} = \gamma V_{gr} \quad (12)$$

The formulation adopted for the calculation of the parameter γ is based on work by Harper et al. (2001), where

$$\begin{aligned} \gamma &= 0.81 & V_{\text{gr}} < 6 \text{ m/s} \\ \gamma &= 0.81 - 2.93 \times 10^{-3}(V_{\text{gr}} - 6) & 6 \leq V_{\text{gr}} < 19.5 \text{ m/s} \\ \gamma &= 0.77 - 4.31 \times 10^{-3}(V_{\text{gr}} - 19.5) & 19.5 \leq V_{\text{gr}} < 45 \text{ m/s} \\ \gamma &= 0.66 & 45 \leq V_{\text{gr}} \text{ m/s} \end{aligned} \quad (13)$$

The inflow angle correction is applied to represent the cross-isobaric flow caused by surface friction and based on the work of Sobey et al. (1977) is given as

$$\begin{aligned} \beta &= 10 \frac{r}{R_1} & r < R_1 \\ \beta &= 75 \frac{r}{R_1} - 65 & R_1 \leq r < 1.2R_1 \\ \beta &= 25 & 1.2R_1 \leq r \end{aligned} \quad (14)$$

3. Input data and quality

3.1. Wind model input parameters

Historical tracks and central pressures of tropical cyclones (Fig. 1) in the North-Eastern Australian region were obtained from the Australian Bureau of Meteorology (BoM) electronic best-track archive (<ftp://ftp.bom.gov.au>). In a very few cases, we have corrected obvious errors in position. None of the values of central pressure were changed. Measurements of eye diameter (used for estimation of radius of maximum winds) in the best-track archive are present only for a small number of points for a small number of cyclones that came within range of weather radar or had an eye that is clearly visible on a satellite image (usually for mature cyclones). Consequently, the best-track archive provides only values of position and central pressure. The remaining parameters in the wind model (i.e. R_1 , R_2 , B_1 , B_2 , p_∞ , θ_{max} , V_s , and θ_s) are determined using a fitting process. This paper describes the process to determine these parameters.

3.2. Wind and pressure data

The measured wind and pressure data set comes from the BoM and contains wind speed, wind direction and atmospheric pressure collected at coastal and offshore automatic weather stations (AWS). The data are of varied quality. One of the greatest benefits of the data set is the large number (more than 20) of meteorological stations in open-water locations such as on coral reefs and cays scattered throughout the Great Barrier Reef (Fig. 2). Unfortunately, the archived electronic data sets for most sites contain measurements only every 3 h and nearly 5 years (1981–1985) of data are unavailable at most offshore sites. For this project, these data were adopted without modification or correction even though the data occasionally exhibited obvious errors. An important long-term goal would be to analyse and correct this data set, but this effort is far too time consuming for the

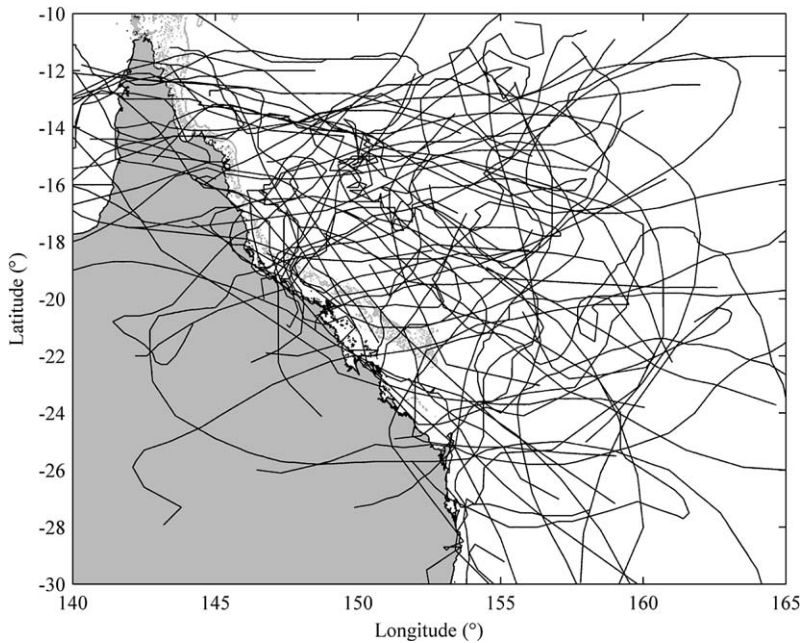


Fig. 1. Tracks of Coral Sea tropical cyclones included in study. Only cyclones that impact the Great Barrier Reef are included.

present project. An optimal wind and pressure data set would comprise measurements in open-water locations measured at 10 m above sea level stored every 10 min (including 10-min mean and the highest 1-min and 3 s gust).

Although most of the land stations are located such that the measurements may not be representative of over-water wind speeds and directions, no correction for this terrain effect was applied to the measured winds. No such correction factors have been derived for the Australian weather stations. Often weather stations that are located on land cannot easily be adjusted for terrain category because complex terrain such as hills and mountains in the local region can significantly alter the wind speeds from their over-water values. However, a project such as the Automated Surface Observing System (Powell, 2002) would be very useful to researchers for model validation studies (e.g. this study, Local Area Prediction System model validations by Tang et al., 1999). The aim of the Automated Surface Observing System project is to determine correction factors to apply to measured wind speeds to give a standard and uniform (across stations) wind measurement.

The present study would also benefit from easy access to data such as paper wind traces, satellite and radar images that are not available online or in electronic format. Unfortunately, access and analysis of such data for all cyclones used in this study would require considerable resources.

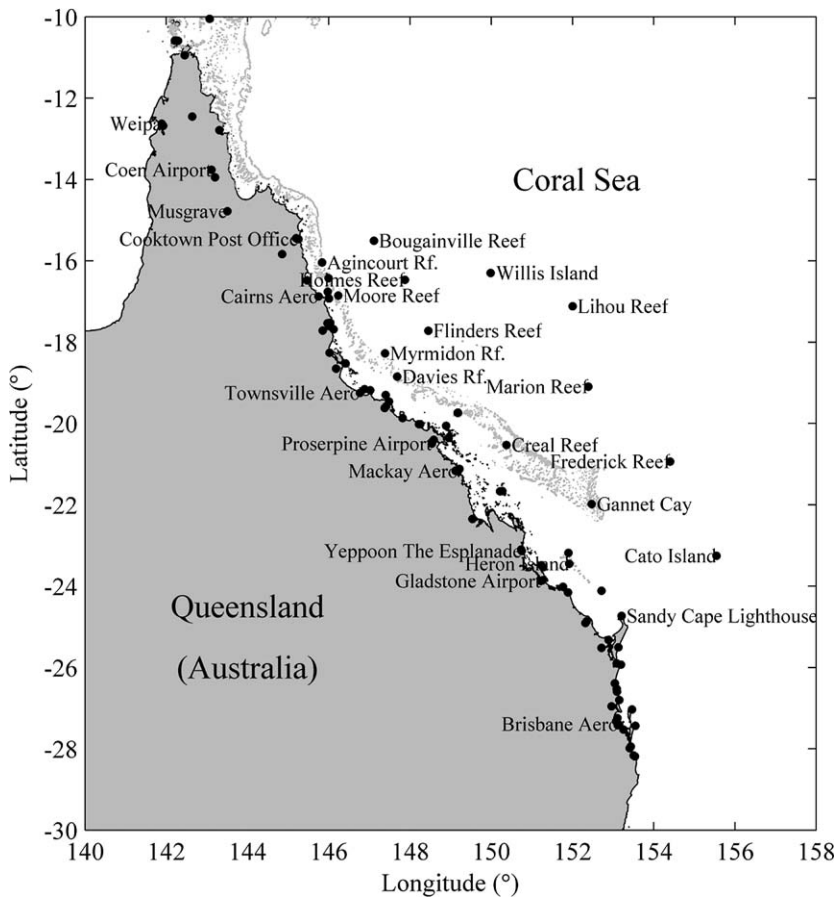


Fig. 2. Locations of meteorological observing stations from which measured wind speed, direction and barometric pressure data have been obtained.

3.3. Calibration of the cyclone wind field

Wind fields have been developed using the combination of a flexible cyclone wind field model, an interactive calibration tool, and all easily accessible data. The calibration process requires modification of wind model input parameters (e.g. radius of maximum winds) until a close match is obtained between modelled and measured data. The measured data comprise wind speed, wind direction and pressure from more than 100 fixed meteorological measurement stations (Fig. 2).

The calibration of the wind fields poses a multi-dimensional optimisation problem that is not easily solved by computer, since mismatches in the degrees of freedom between the number of inputs and number of outputs are common. In addition, data gaps, irregularity in timing, asynchronous data and obvious data

errors require experienced visual scrutiny to subjectively discern the “goodness” of fit.

The Marine Modelling Unit (MMU) has developed a graphical user interface tool, CyCal, to facilitate this optimisation process. CyCal is a MATLAB program that allows interactive calibration of tropical cyclone wind fields. This computer program is similar in concept to that described in Cox and Cardone (2000) with the exception that the underlying model is parametric. The tool has a built-in wind field model to compute the wind velocity and pressure field on-the-fly during interactive changes of the cyclone parameters. CyCal requires time series of cyclone model parameters and time series of wind speed and direction and pressure data as its inputs.

CyCal has the facility to display time series of wind speed, direction and pressure data at many measurement stations, compute error and regression statistics and present plots of modelled versus measured data. These graphics and statistics are calculated immediately and enable the user to visualise the effect of each change of input parameters. This creates a strong and fast feedback loop for iterating until a close match is obtained between modelled and measured wind speed, direction and pressure data at multiple locations.

In addition to calibrating 64 historical tropical cyclones of the Coral Sea region presented here, CyCal and CycWind have been used to calibrate 15 tropical cyclones in the North-Western Australian region with very good success, among them the very severe Olivia, Orson and Vance (McConochie et al., 2002).

4. Results for individual historical cyclones

Results for three historical cyclones, Simon, Justin, and Joy, will be presented. These cyclones have been chosen to illustrate some of the more common issues that arise during the model fitting process.

Three statistics are used in the following discussions; mean error (e_m) (Eq. (15)) and root mean square error (e_{rms}) (Eq. (16)) and scatter index (s) (Eq. (17)). To provide stable and meaningful statistics, only measurements taken while the cyclone was closer than 300 km are used in the calculation of these statistics. The statistics are calculated using

$$e_m = \frac{1}{N} \sum_{i=1}^N x_o - x_e \quad (15)$$

$$e_{rms} = \frac{1}{N} \sum_{i=1}^N (x_o - x_e)^2 \quad (16)$$

$$s = \frac{e_{rms}}{\bar{x}_e} \quad (17)$$

where x_o is the modelled value, x_e is the measured value, $\bar{x}_e$ is the mean measured value, and N is the number of data points.

4.1. T.C. Simon

T.C. Simon was a significant cyclone that affected the Capricorn coast region of Queensland (Fig. 3). It tracked south-westerly after forming in the middle of the Coral Sea and steadily intensified until reaching maximum intensity of 950 hPa near the coastal town of Yeppoon. Simon then turned south-eastwards tracking parallel to the coast and began filling. It passed within 100 km of five automatic weather stations (Marion Reef, Cape Capricorn Lighthouse, Heron Island, Lady Elliot Island, and Sandy Cape Lighthouse). Measurements taken by stations as the cyclone passed play an important role in selecting R_1 and B_1 . If a cyclone does not pass close to a measurement station, the fitting process for these two important parameters becomes more subjective.

The wind model input parameters were adjusted until the modelled values of wind speed, direction and pressure achieved the best match with the measured data. Of the more than 100 measurement sites, only 22 stations measured data suitable for use in calibrating the wind field of T.C. Simon. For each of the 64 cyclones, the number of stations used in the calibration varied depending on the data availability and track of the storm.

Time series comparisons of measured and modelled wind speed and pressure are shown in Fig. 4 for the five stations that were within 100 km of the track of T.C. Simon sometime during the storm. One important issue with the measured data is

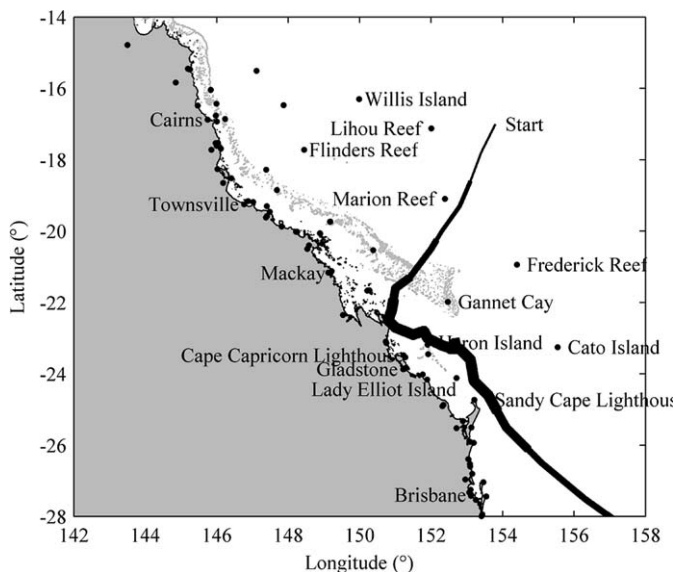


Fig. 3. Track of tropical cyclone Simon occurred February 1980 with minimum central pressure of 950 hPa. Thickness of the track indicates central pressure: — $p_0 \geq 990$ hPa; — $989 > p_0 \geq 980$ hPa; — $970 > p_0 \geq 969$ hPa; — $960 > p_0 \geq 959$ hPa; — $p_0 \leq 959$ hPa.

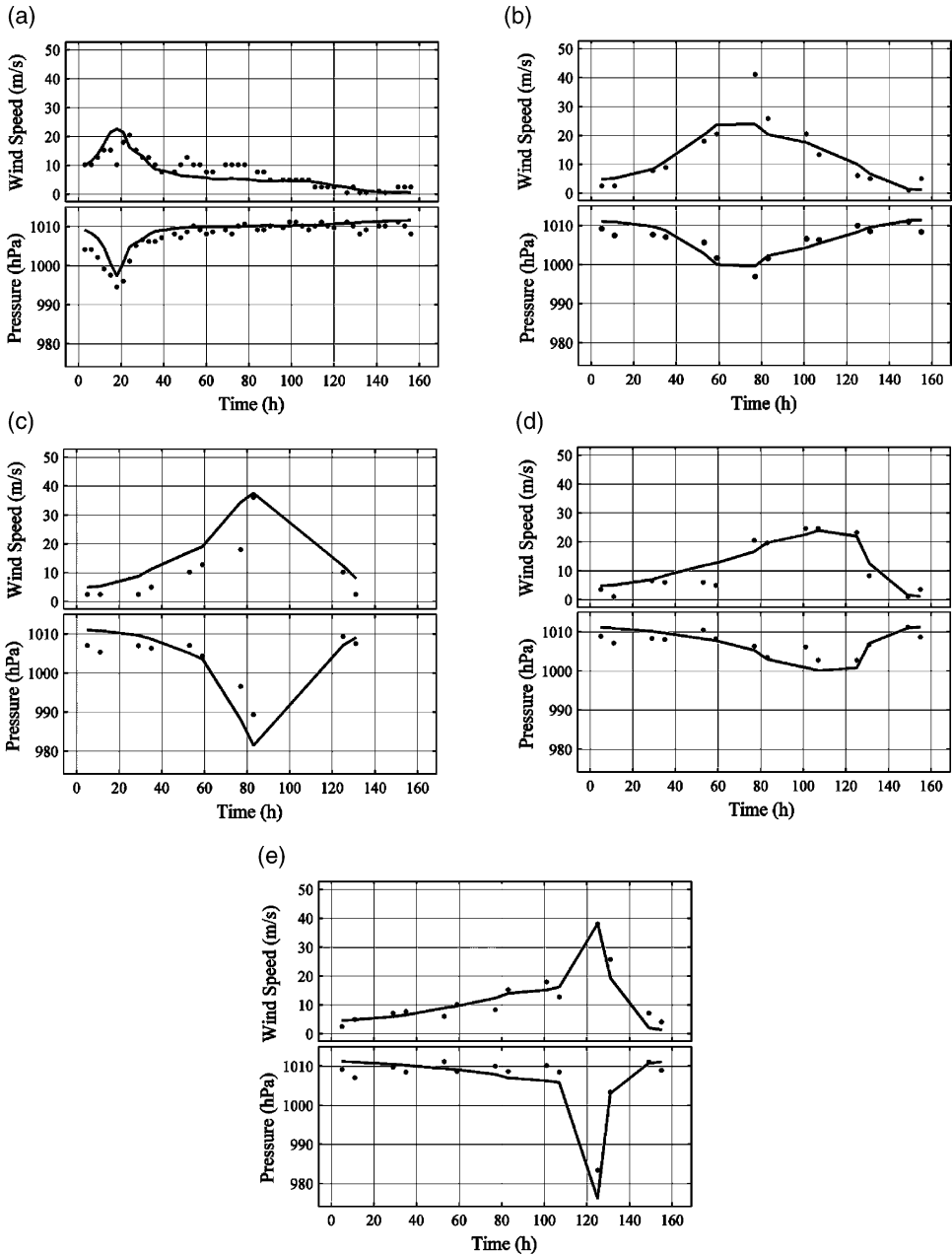


Fig. 4. Comparison of modelled (—) and measured (●) time series of wind speed and pressure during T.C. Simon.

evident in the comparisons; the measurements are often irregular and intermittent (e.g. Heron Island Research Station). Note also that only very few of the measure-

ments in Fig. 4 are taken when the centre of the cyclone was within 100 km of the station, in this case, fewer than 40 individual samples.

A good fit is obtained at Sandy Cape Lighthouse (Fig. 4e) with the peak wind speed matching very closely (less than 1 m/s difference). The mean error (e_m) is less than 0.5 m/s and root mean square error (e_{rms}) is less than 4 m/s. Similarly good fits are obtained at Marion Reef and Lady Elliot Island. However, results at Cape Capricorn and Heron Island indicate the model under-estimates and over-estimates the wind speed, respectively. This is a common occurrence during calibrations and the best fit is often a subjective compromise (based on the experience of the user) sometimes favouring one station over another.

Fig. 5 shows scatter diagrams of modelled versus measured (synchronous) for wind speed, wind direction and pressure.

4.2. T.C. Justin

T.C. Justin is well known mostly because of its massive spatial extent (radius greater than 150 km) during the first half of its life and its longevity lasting nearly 19 days (Fig. 6). When declared, it was positioned in the middle of the Coral Sea where it lingered for nearly 6 days. It then began to track north and then north-east reaching peak intensity of 955 hPa near 155° E, 12° S (after reducing in spatial extent). It then tracked steadily south-west until crossing the coast just north of Cairns and recurving over land towards Townsville.

Justin was difficult to calibrate because of its size and interaction with a mid-latitude high-pressure system (1020 hPa) which are typical during the summer months (Tapper and Hurry, 1993). As a result, it was important to use each component (i.e. primary and secondary vortex formulations and the synoptic wind blending formulation) of CycWind to properly calibrate Justin. The match achieved for wind speeds, direction and MSL pressure was excellent (Figs. 7 and 8). The only significant deviation of the model wind speed estimates from measured wind speeds is observed in the first 5 days for stations located in a region to the

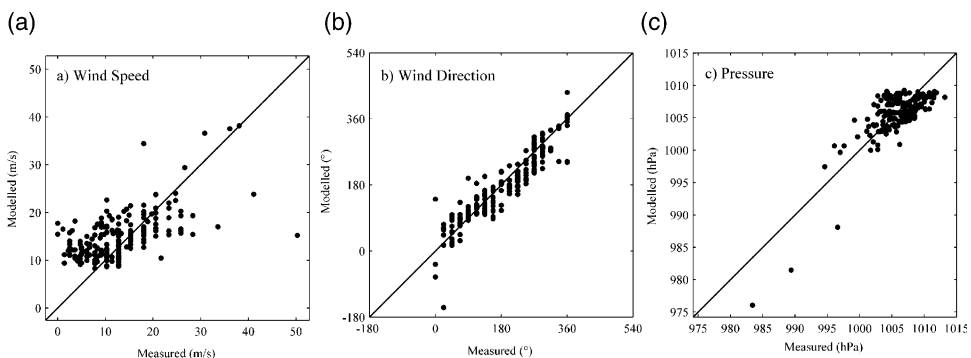


Fig. 5. Comparison of modelled and measured (a) wind speed, (b) wind direction and (c) mean sea level pressure for the calibration of T.C. Simon.

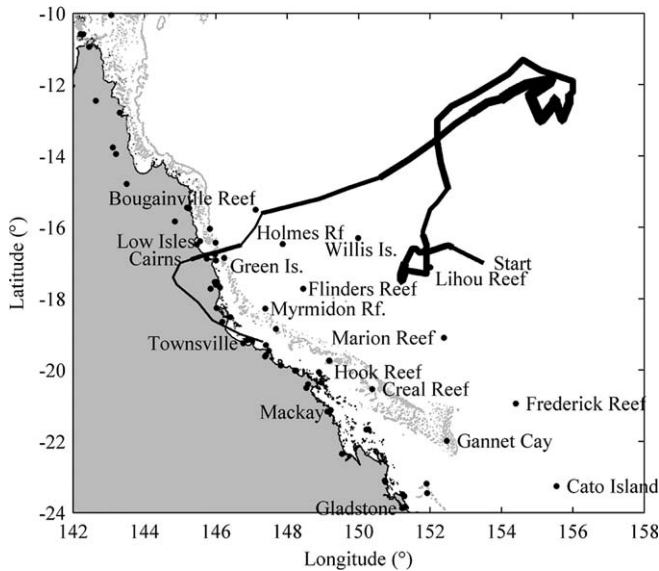


Fig. 6. Track of tropical cyclone Justin of March 1997 with minimum central pressure of 955 hPa. Thickness of the track indicates bands of central pressure as in Fig. 3.

west to north-west of the cyclone centre. Stations in this region; Bougainville Reef, Low Isles, Holmes Reef and Myrmidon (Fig. 8) recorded wind speeds significantly lower than the model estimates, but for stations elsewhere (Fig. 7), model estimates matched excellently. These lower wind speeds to the west and north-west are likely a result of the beta-effect distortion (Holland, 1984) and since this effect is not represented in this wind model, the wind speeds are over-estimated in this region. The diurnal fluctuations in pressure due to day night variations in temperature cause the high frequency fluctuations in the pressures shown in Figs. 7 and 8.

The scatter plot of modelled versus measured wind speeds (Fig. 9) and modelled versus measured MSL pressures clearly illustrate the veracity of the model. The mean error, e_m , for wind speed and pressure is less than 3 m/s and 4 hPa, respectively and e_{rms} of wind speed and pressure is less than 5 m/s and 5 hPa, respectively.

4.3. T.C. Joy

T.C. Joy (1990) developed in the north-eastern section of the Coral Sea and moved steadily west then south-west, slowly intensifying reaching maximum intensity (940 hPa) just east of Low Isles (Fig. 10). Joy exhibited a concentric-eye structure (Willoughby et al., 1982) while observed by Cairns radar (Callaghan, personal communication). Joy then moved slowly southwards finally turning towards land (south-west) and crossing the coast near Ayr (just south of Townsville).

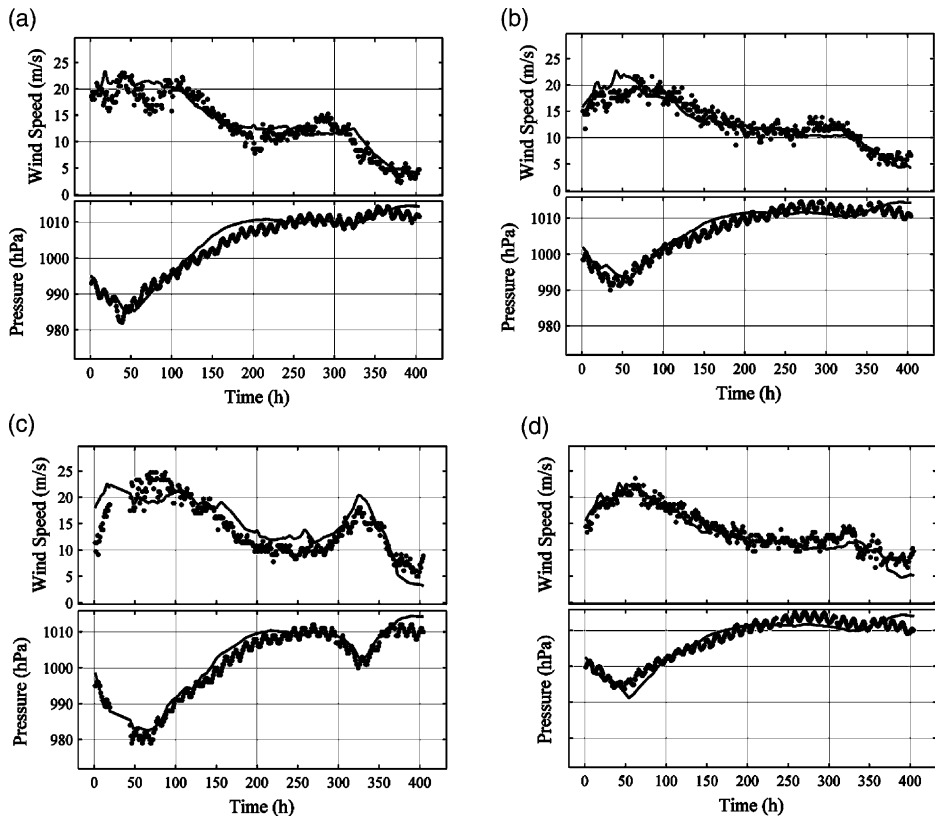


Fig. 7. Comparison of modelled (—) and measured (●) time series of wind speed and pressure during T.C. Justin.

Generally, the match of the model to the measured data is good, although variable, with e_{rms} for wind speed and pressure below 3 m/s and 4 hPa, respectively. The scatter plots of modelled versus measured wind speed and pressure (Fig. 11) illustrates the fit of the model. Low Isles station measured wind speeds greater than 25 m/s during the passage of Joy (within 120 km) and the model matches very well. In contrast, later on, as Joy begins to move towards the coast, the model estimates at Myrmidon Reef are less successful (Fig. 12). The under-estimated wind speed between 150 and 200 h and the poor match in the pressure remains unexplained. It was not possible to match this feature of the measured time series.

The Cairns meteorological station measured wind speeds that are significantly lower than modelled as opposed to Low Isles for which the model matched excellently. But, these stations are only 54 km apart, and are nearly the same distance from the cyclone (123 km for Cairns and 134 km for Low Isles). We believe that most of this difference is attributable to over land effects. As previously discussed,

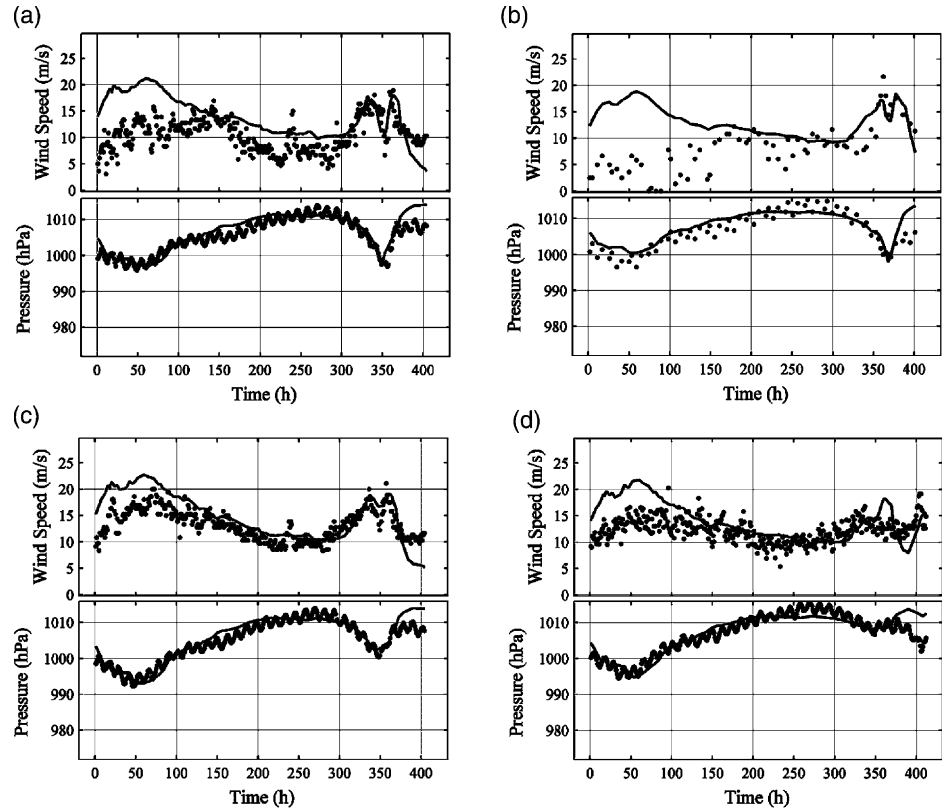


Fig. 8. Comparison of modelled (—) and measured (●) time series of wind speed and pressure during T.C. Justin.

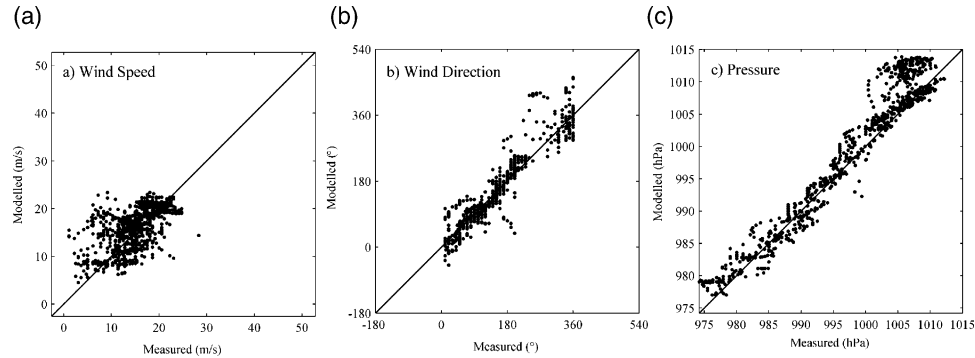


Fig. 9. Comparison of modelled and measured (a) wind speed, (b) wind direction and (c) mean sea level pressure for the calibration of T.C. Justin.

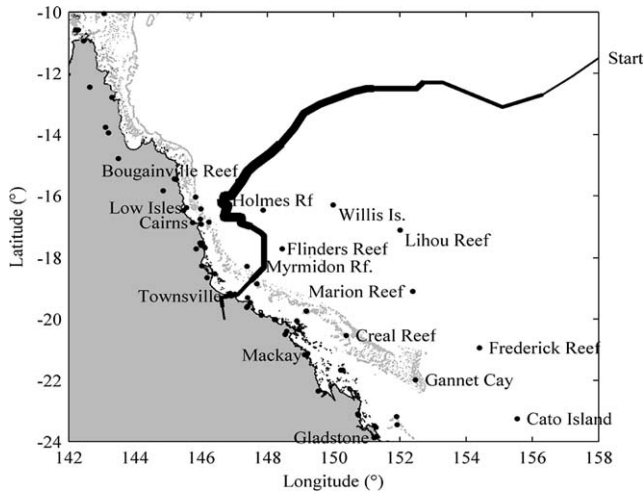


Fig. 10. Track of tropical cyclone Joy of December 1990 with minimum central pressure of 940 hPa. Thickness of the track indicates bands of central pressure as in Fig. 3.

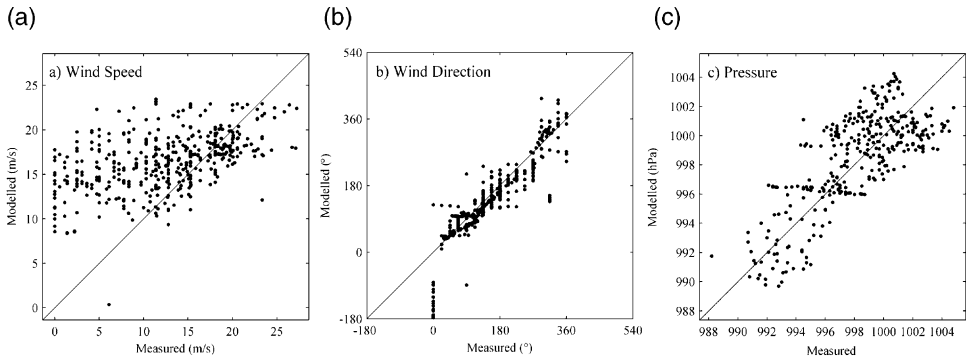


Fig. 11. Comparison of modelled and measured (a) wind speed, (b) wind direction and (c) mean sea level for the calibration of T.C. Joy.

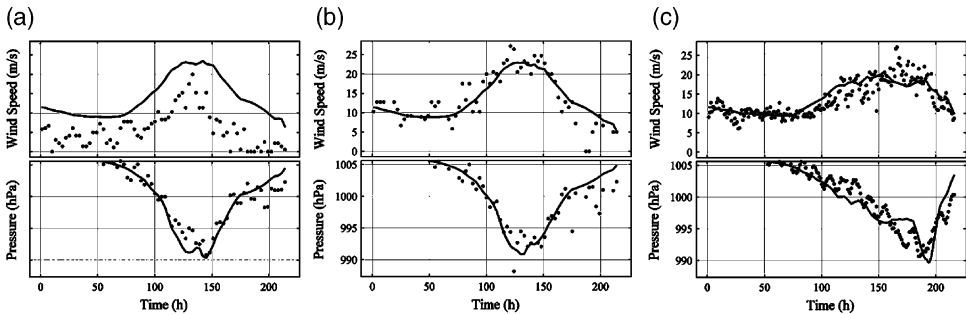


Fig. 12. Comparison of modelled (solid line) and measured (markers, x) time series of wind speed and pressure during T.C. Joy.

the model does not account for the frictional and topological effects of land on wind speeds and direction.

5. General results of calibrations of historical cyclones

5.1. Statistics of fit

An overview of the performance of the calibrated wind fields for the 64 historical storms is given in this section. At each measurement time, the statistics are calculated including data from all stations which, at that time, are within 300 km of the centre of the storm.

Fig. 13 gives histograms of errors e_m and e_{rms} for wind speed, direction and MSL pressure. Errors have been compiled for 64 cyclones (Appendix A, Fig. 1). The average (μ) and standard deviation (σ) of the errors is given in the top-right corner of each figure. The left-hand plots (Fig. 13a, c and e) are for e_m and the right-hand side (Fig. 13b, d and f) are for e_{rms} .

It is clear from the figures of mean error (Fig. 13a, c and e) that the calibrated wind fields are unbiased for wind direction ($\mu = -2.2^\circ$) and pressure ($\mu = -0.2$ hPa) as these results are within the expected precision of the data collection. However, the results are biased slightly high for wind speed ($\mu = +1.1$ m/s), although the mode of the e_m for wind speed is 0 m/s. The corresponding e_{rms} (Fig. 13b, d and f) are also very good with averages 5.1 m/s, 33° and 2.4 hPa for wind speed, direction and MSL pressure, respectively. The mode of e_{rms} for wind speed is excellent at 3 m/s. The average scatter index (s) of wind speed is 49% with a mode of 20% (not shown); also a good result.

The above statistics included land-based stations. To understand the impact that land based weather stations, and the lack of terrain category correction have on the statistics, we compare the statistics with the land-based stations removed from the analysis. Table 1 gives statistics with and without land-based stations. As expected, the removal of the land-based stations from the calculation of the statistics improves all results, particularly the wind speed and direction. Note that the wind speed from the model is not biased if only offshore stations are considered. The scatter index for wind speed using only offshore stations is $s = 0.36$ compared to $s = 0.49$ for all stations; a significant difference and an excellent result.

Although the statistics are very good, it is important to realise the effect of the wind field on the oceanic processes it is being applied to fields of wave, storm surge and water currents. While some of the effects from changes of the radius of maximum wind and forward speed on the generation of waves have been examined by Young (1988), the shape parameter, B_0 (B_1 in model) is also potentially significant. This parameter is the most difficult to tune, but may have a significant impact on the modelled wave field and the maximum significant wave height.

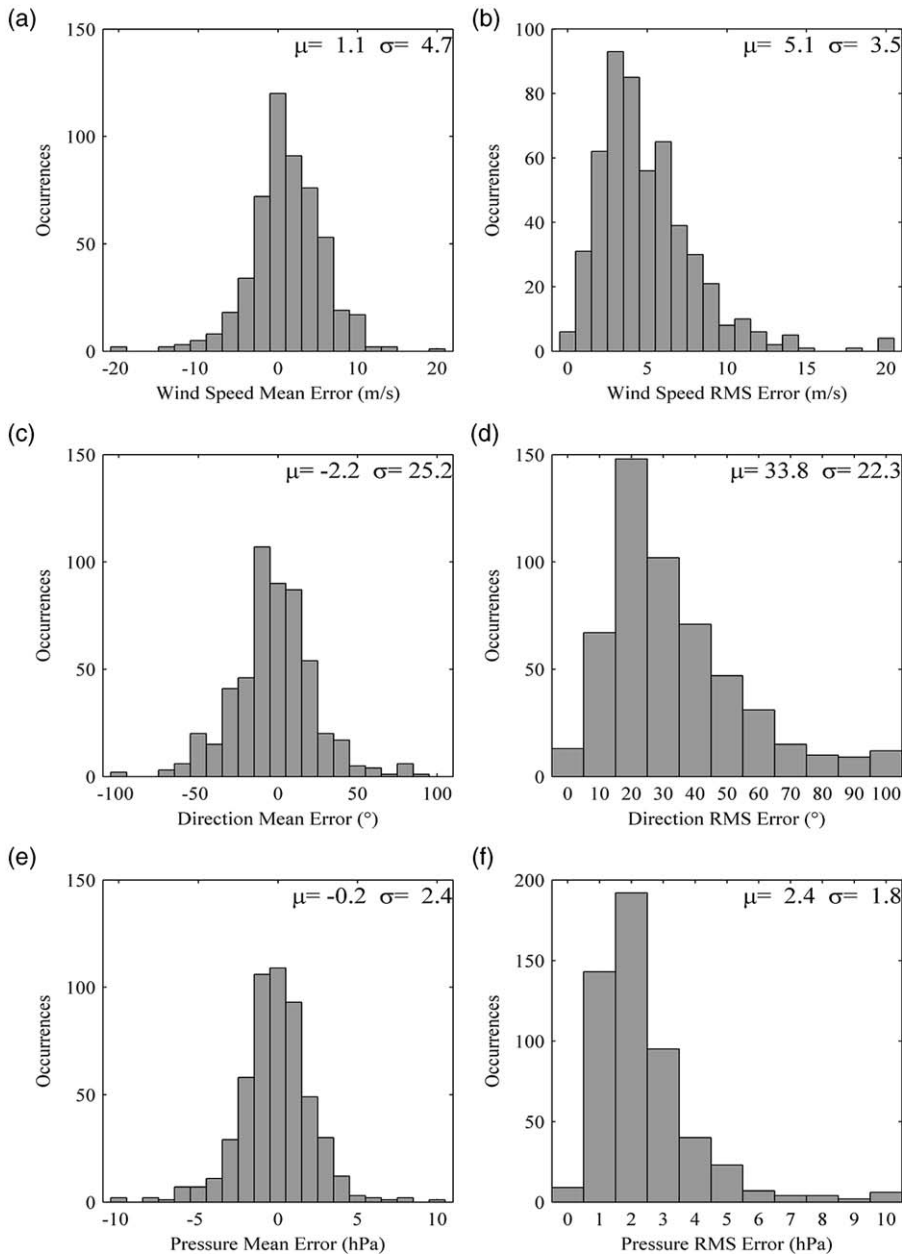


Fig. 13. Distributions of errors for wind speed, wind direction and MSL pressure resulting from the calibration procedure. Includes 64 historical tropical cyclones listed in Appendix A.

Table 1

A comparison of calibration statistics between offshore and land stations (ALL) and offshore stations (OFFSHORE)

	ALL stations		OFFSHORE stations	
	Mean error	RMS error	Mean error	RMS error
Wind speed (m/s)	1.1	5.1	0.0	4.7
Wind direction (°)	−2.2	33.8	0.3	32.4
MSL pressure (hPa)	−0.2	2.4	0.0	2.4

6. Maximum wind speeds

The wind model used in this study, based on the momentum equation, produces an ensemble mean wind speed. For a stationary process, the ensemble mean is equivalent to the temporal mean averaged over 30–60 min for atmospheric winds (Stull, 1988). However, since the measured winds, which are 10-min mean, have been used throughout the fitting process, with no bias, we take the model winds as being equivalent to a 10-min mean wind speed.

The spatial distribution of modelled maximum of the 10-min mean wind speeds determined for 64 historical cyclones is given in Fig. 14. The figure shows the maximum of the modelled 10-min mean wind speed at each calculation point for the ensemble of 64 tropical cyclones. It is clear that the distribution of cyclones is sparse with large sections of the coastline and Great Barrier Reef unaffected by strong cyclonic winds in the past 30 years. This sparse distribution has implication

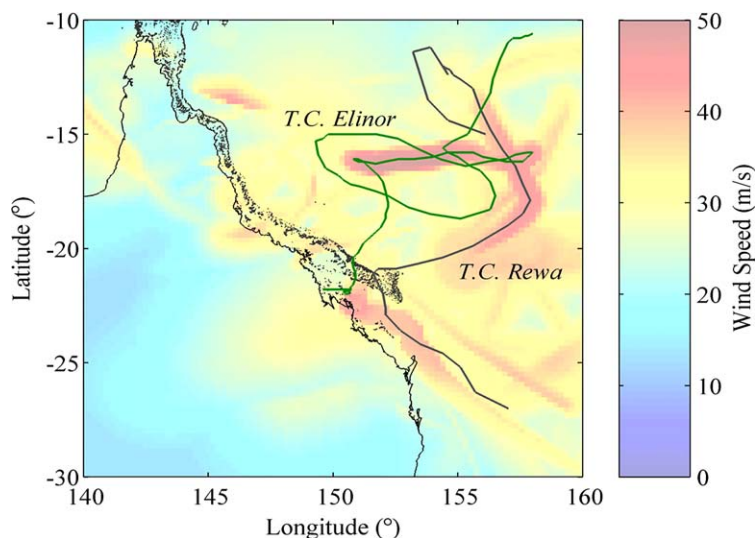


Fig. 14. An image illustrating the maximum of the 10-min mean wind speed at each computational point from 64 historical tropical cyclones.

for engineers interested in developing statistics of tropical cyclone winds for use in design of structures. It is clear that 30 years of data are insufficient to establish, for example, the 100-year return period wind speed. However, a method of establishing the climate of tropical cyclones in the Great Barrier Reef region has been developed by James and Mason (1999) using synthetically (computer) generated cyclones. The synthetic cyclone model is an auto-regressive time series model that calculates time-series of cyclone positions and central MSL pressures. Historical cyclone tracks and pressures are used to calibrate the synthetic model. With this model, it is possible to generate thousands of synthetic cyclones for the Coral Sea region, representing thousands of years thus overcoming the problem of sparseness of historical cyclones.

Two cyclones prominent in the spatial plot of maximum wind speed (Fig. 14) are T.C. Elinor and T.C. Rewa. Elinor reached a minimum MSL pressure of 935 hPa, not the most intense cyclone, but the calibration parameter of B_1 needed to match the data was very high ($B_0 = 8.45$) which gives a very peaked wind field profile with a very high maximum wind speed (≈ 50 m/s). Rewa reached a minimum MSL pressure of 920 hPa, coupled with a 35 km R_1 , developed high mean wind speeds of 44 m/s. Although these cyclones did not directly impact the Queensland coast during these maximum wind periods, they were included because of the potential for significant wave generation that may impact the GBR.

The latitudinal distribution of maximum wind speeds due to cyclones is shown in Fig. 15. Each vertical colour bar represents one cyclone with the maximum wind

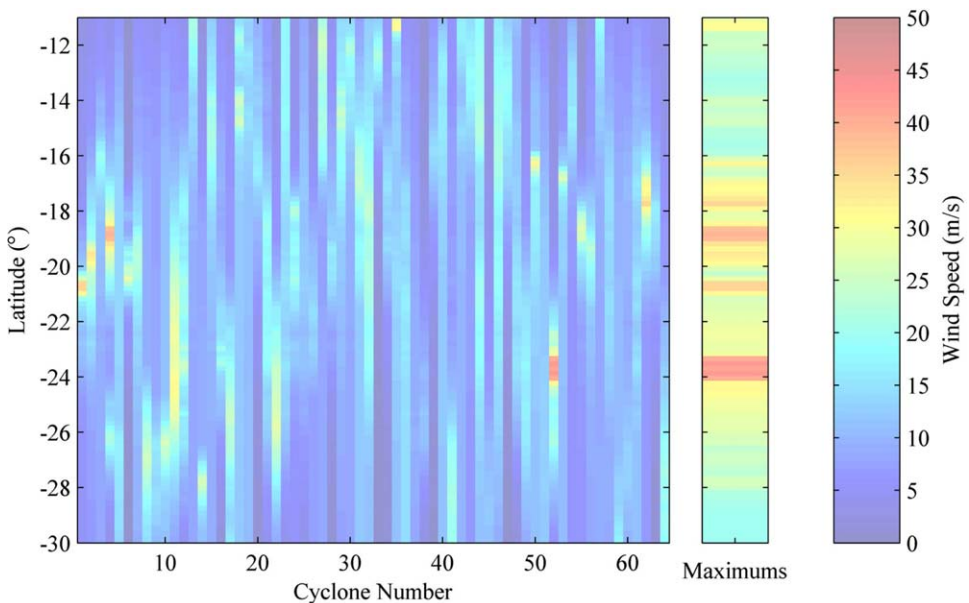


Fig. 15. Maximum 10-min mean wind speeds along the Queensland East Coast for each of the 64 cyclones modelled (see Appendix A for storm number).

speed at each coastal point over all of the cyclones indicated in the bar adjacent to the main axes. Each of the numbers indicated on the x -axis corresponds to the cyclones listed in Appendix A. The result is as expected, with weaker winds speeds at higher ($25\text{--}30^\circ$) and lower latitudes ($10\text{--}15^\circ$) and stronger wind speeds near latitude 20° S. Again, it is evident that large sections of the coast have been (for this 30 year period) unaffected by cyclone forced wind speeds of a significant magnitude.

It is clear that cyclone number 52 (Fig. 15), T.C. Simon (935 hPa) contributed significantly to the maximum wind speeds near Hervey Bay (latitude $\approx -23^\circ$). During Simon, 10-min mean wind speeds measured at the Cape Capricorn Lighthouse exceeded 40 m/s and at two other weather stations exceeded 35 m/s. These are among the *highest* 10-min mean wind speeds measured during east coast tropical cyclones.

7. Distribution of wind field input parameters

A most useful outcome of this calibration exercise is the distribution of values of the cyclone wind field parameters, particularly radius of maximum winds and B_0 . The distribution of values of radius to maximum winds is critical for applications that develop statistics of extreme levels of storm surges, waves heights and wind speeds using Monte Carlo simulation methods. In past studies (e.g. Gomes and Vickery, 1976; Harper, 1999), researchers and engineers have adapted distributions from other tropical cyclone basins, since insufficient measurements of R_1 exist for the Coral Sea cyclone basin. Note that B_1 is usually held constant for these studies. The distribution typically adopted for R_1 is the lognormal distribution (Georgiou, 1985). Hardy et al., 2003 allowed the lognormal distribution to change in time with changing values of central pressure and randomly chose a relative position in the distribution that remained constant during the storm.

Fig. 16 presents the distributions of the parameter values resulting from this calibration study. The distribution of radius of maximum wind (Fig. 16a) values is generally as expected, with a mode of 30 km and most values between 10 and 60 km. An interesting feature is the occurrence of 12 values of R_1 greater than 100 km. These values cause the mean of R_1 to be higher than expected at 57 km. Also, note the lack of any values in the 5–15 km range. This is unexpected but probably a result of the sparseness of the measurement stations. For the calibration to be reliable, we believe that: (i) a cyclone should pass within about $3R_1$ of a measurement station; and (ii) the measurement station should take samples at a sufficient temporal rate (i.e. at least one every 10–40 min). This situation does not occur often especially for smaller storms.

Note a value of 30 km for R_1 was used as a starting point in the calibrations. We expect this to cause some bias, since sometimes, there are insufficient data to justify changing R_1 from this default value. Nevertheless, the lognormal distribution that has been fitted to the distribution of calibrated R_1 values (Fig. 16a) appears sensible with mean and standard deviation of 57 and 39 km, respectively.

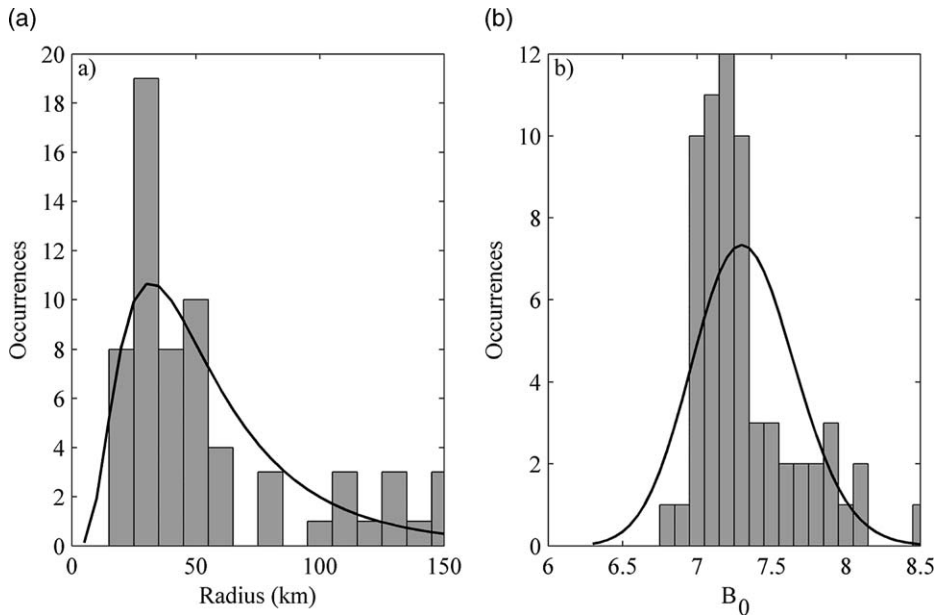


Fig. 16. Histograms of (a) radius of maximum winds and (b) Holland B_0 parameter resulting from the calibration of 64 historical tropical cyclones and the best-fit lognormal distributions.

The distribution of B_0 (Fig. 16b) with few high values (more peaked) appears reasonable given the general consensus that more intense cyclones are more peaked and few intense cyclones are in the data. As for R_1 , if the cyclone does not pass close to a measurement site, the peak is very difficult to tune. Note that since the starting value for all the cyclones was 7.3, a bias in these values would likely favour this value. This parameter may be an important one in tropical cyclone wave modelling studies and poor selection of the value of B_1 could be responsible for poor wave model results, even where one comparison of modelled wind speed and direction appears to compare well with measurements. Comparisons with multiple measurement stations are required to obtain confidence in the shape of the cyclone wind field.

8. Examples of use of calibrated wind fields

The calibrated wind field model, CycWind has been used to produce simulations of storm surge (Harper et al., 2001) and waves (Hardy et al., 2000) during historical tropical cyclones in the GBR region, and also for synthetic storms (Hardy et al., 2003). Good quality measurements of large storm surges and waves are rare due to the lack of measurement stations and the infrequency of strong tropical cyclones. However, an occurrence of both surge and waves from relatively strong storms has been recorded. Fig. 17 shows modelled versus measured results for storm surge

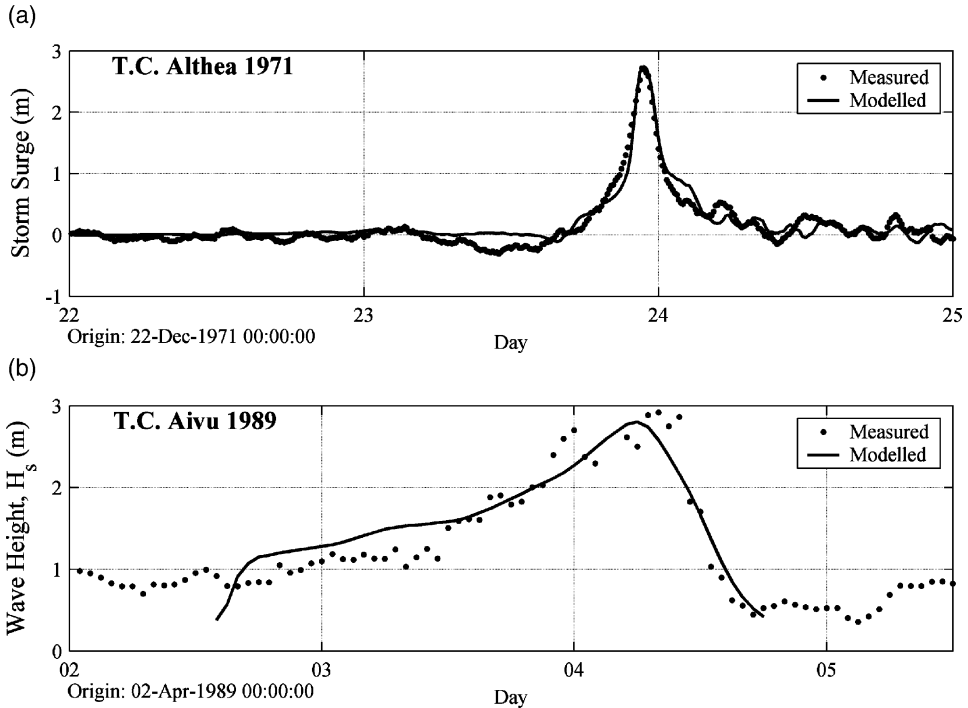


Fig. 17. Examples of comparisons of model estimated storm surge and wave height with measurements under tropical cyclone conditions. (a) Time series of storm surge generated under T.C. Althea 1971; (b) significant wave height within the GBR reef matrix under T.C. Aivu 1989.

(Fig. 17a) during T.C. Althea and for waves (Fig. 17b) during T.C. Aivu. The storm surge is the largest recorded at the Townsville tide gauge. The wave buoy did not record the largest waves during Aivu, but the location of the buoy inside the matrix of reefs of the GBR presents a difficult modelling challenge for the combined wind and wave numerical models.

The results shown in Fig. 17 are excellent. It is astounding that the complexity of the moving wind field of a tropical cyclone can be modelled effectively by a parameterised wind field model and that wind field can then drive storm surge and wave models to produce accurate results.

9. Conclusion

Wind fields of 64 historical tropical cyclones have been developed using a parametric wind field model and an interactive computer tool. The result from the study is 64 cyclone wind fields spanning 33 years (1969–2002) of cyclone activity along the Queensland East Coast. These wind fields are suitable to be applied to applications such as wave, storm surge and circulation modelling. The parametric

double vortex wind field model is based on [Holland \(1980\)](#). The process used all easily accessible meteorological data, including wind speeds, directions and pressures to determine the best-fit wind fields.

It is clear that data measured by the sparse AWS is, in some cases, insufficient alone to develop accurate wind fields of tropical cyclones in the Queensland coastal region. However, in most cases, the parametric model adopted in this study has been shown to produce unbiased wind fields using the weather station data and synoptic charts alone with an acceptable level of wind speed scatter.

Further work in several areas is still needed: the characterisation of the exposure of each of the measurement stations to incorporate into the calibration tool; contributions of data from unpublished studies and sources; improvement in the wind field representation to possibly include the effect of super-gradient winds (from recent work by [Kepert, 2001](#) and [Mallett, 2000](#)) and; account of asymmetries of the wind field other than due to the forward motion of the cyclone.

In addition, not enough is known about potential temporal changes in the storm radius of maximum winds and Holland *B* parameters; how quickly can a cyclone change radius or shape? During the calibration, we can introduce a change in the radius or Holland *B* in time that should, but may not, satisfy the conservation of energy and angular momentum within the cyclone. Measurements of the temporal variations of these variables are not yet available for a reasonable cost, but with the advent of autonomous meteorological aircraft, such as the Aerosonde ([Holland, 2002](#)) these data may become available and would be of great benefit.

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Appendix A. List of cyclones included in study

Cyclone number	Start date	End date	Minimum central pressure	Cyclone name	MSL within reef
1	02-Jan-1970	18-Jan-1970	962	Ada	963
2	01-Apr-1989	05-Apr-1989	935	Aivu	950
3	29-Jan-1976	09-Feb-1976	989	Alan	996
4	19-Dec-1971	29-Dec-1971	952	Althea	952
5	13-Feb-1976	22-Feb-1976	972	Beth	994
6	26-Jan-1996	29-Jan-1996	965	Celeste	965

7	21-Feb-1988	01-Mar-1988	972	Charlie	985
8	09-Feb-1981	15-Feb-1981	975	Cliff	985
9	25-Feb-1976	04-Mar-1976	954	Colin	954
10	05-Feb-1972	13-Feb-1972	959	Daisy	968
11	13-Jan-1976	21-Jan-1976	961	David	963
12	03-Mar-1976	06-Mar-1976	988	Dawn	988
13	15-Feb-1996	18-Feb-1996	990	Dennis	995
14	10-Feb-1971	17-Feb-1971	990	Dora	993
15	08-Feb-1981	13-Feb-1981	980	Eddie	985
16	10-Feb-1983	03-Mar-1983	935	Elinor	981
17	27-Mar-1972	04-Apr-1972	942	Emily	961
18	08-Mar-1996	13-Mar-1996	982	Ethel	982
19	10-Apr-1972	24-Apr-1972	990	Faith	1002
20	13-Dec-1989	20-Dec-1989	975	Felicity	999
21	16-Feb-1971	28-Feb-1971	960	Fiona	995
22	09-Mar-1992	17-Mar-1992	950	Fran	980
23	24-Feb-1981	07-Mar-1981	962	Freda	990
24	10-Feb-1971	16-Feb-1971	983	Gertie	983
25	15-Jan-1975	19-Jan-1975	976	Gloria	994
26	08-Jan-1979	11-Jan-1979	988	Gordon	1001
27	08-Jan-1979	12-Jan-1979	986	Greta	990
28	23-Feb-1997	24-Feb-1997	994	Ita	994
29	16-Mar-1990	26-Mar-1990	965	Ivor	970
30	05-Mar-1984	09-Mar-1984	976	Jim	980
31	18-Dec-1990	27-Dec-1990	940	Joy	983
32	06-Mar-1997	23-Mar-1997	955	Justin	993
33	16-Mar-1984	23-Mar-1984	920	Kathy	990
34	29-Jan-1977	31-Jan-1977	992	Keith	992
35	24-Feb-1991	05-Mar-1991	980	Kelvin	999
36	12-Feb-1979	04-Mar-1979	945	Kerry	994
37	04-Apr-1984	07-Apr-1984	992	Lance	995
38	13-Mar-1971	19-Mar-1971	980	Lena	988
39	21-Apr-1986	27-Apr-1986	970	Manu	997
40	05-May-1989	10-May-1989	990	Meena	1000
41	28-Jan-1990	04-Feb-1990	975	Nancy	980
42	20-Mar-1998	26-Mar-1998	990	Nathan	990
43	23-Dec-1992	02-Jan-1993	960	Nina	995
44	06-Mar-1977	10-Mar-1977	984	Otto	987
45	03-Feb-1974	06-Feb-1974	965	Pam	
46	02-Jan-1980	08-Jan-1980	983	Paul	992
47	29-Dec-1978	03-Jan-1979	980	Peter	992
48	18-Feb-1985	24-Feb-1985	986	Pierre	986
49	28-Dec-1993	21-Jan-1994	920	Rewa	975
50	09-Feb-1999	21-Feb-1999	970	Rona	970

51	11-Feb-1980	18-Feb-1980	980	Ruth	n/a
52	21-Feb-1980	28-Feb-1980	950	Simon	950
53	27-Feb-2000	11-Mar-2000	975	Steve	980
54	27-Mar-1985	01-Apr-1985	982	Tanya	990
55	01-Apr-2000	02-Apr-2000	980	Tessi	990
56	14-Dec-1973	22-Dec-1973	988	Una	989
57	10-Feb-1970	19-Feb-1970	990	Unnamed	994
58	17-Jan-1974	21-Jan-1974	986	Vera	996
59	03-Mar-1995	08-Mar-1995	960	Violet	990
60	20-Jan-1974	25-Jan-1974	997	Wanda	997
61	25-Apr-1976	28-Apr-1976	970	Watorea	980
62	27-Jan-1986	05-Feb-1986	957	Winifred	961
63	08-Feb-1974	11-Feb-1974	995	Yvonne	995
64	06-Mar-1974	14-Mar-1974	968	Zoe	982

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